

CSE 3112 Midterm 2

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I. QUESTION (20 POINTS)

Find the order of growth for solutions of the following recurrences.

- a) $T(n) = 8T(n/2) + n^2$, $T(1) = 1$
- b) $T(n) = 8T(n/2) + n^3$, $T(1) = 1$
- c) $T(n) = 8T(n/2) + n^4$, $T(1) = 1$

II. QUESTION (20 POINTS)

- a) Write a divide-and-conquer algorithm for the exponentiation problem of computing a^n where n is a positive integer.
- b) What is the time complexity of the algorithm?

III. QUESTION (20 POINTS)

- a) Write a dynamic programming algorithm for the exponentiation problem of computing a^n , where n is a positive integer.
Hint: Store $a^1, a^2, a^4, a^8, a^{16}, \dots$ in a dynamic programming table. Use the binary representation of n .
- b) What is the time complexity of the algorithm?

IV. QUESTION (20 POINTS)

Find the square root of an integer N using divide and conquer technique which run in $\mathcal{O}(\log N)$ time. Assume that, the square root of N is an integer. You are not allowed to use sqrt function.

V. QUESTION (20 POINTS)

Given an array A containing (negative or positive) integers, determine the maximum range sum of A in $\mathcal{O}(N^2)$ time. In other words, find the maximum sum considering any two indices i and j in array A .

Hint: Pre-process the array A to get the dynamic programming table which contains sums from 0 to i . Write the algorithm using the table.